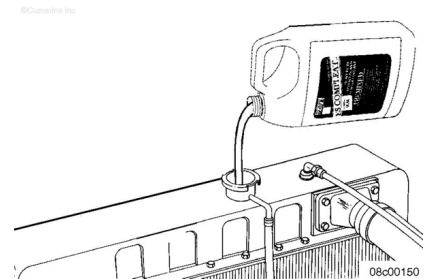


## Trainings ▾

### Fully Formulated Coolant/Antifreeze

#### ⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



**Note :** Low-silicate antifreeze meeting ASTM D4985 is inadequate for these extended service intervals of 80,467 to 241,402 km [50,000 to 150,000 miles].

Cummins Inc. recommends using either a 50/50 mixture of good-quality water and fully formulated antifreeze, or fully formulated coolant when filling the cooling system. The fully formulated antifreeze or coolant **must** meet CES14603 specifications.

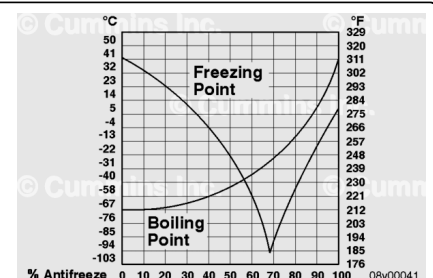
Good-quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.

©Cummins Inc.

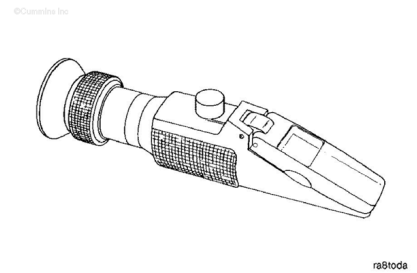
#### Water Quality

|                              |                                                             |
|------------------------------|-------------------------------------------------------------|
| Calcium Magnesium (Hardness) | Maximum 170 ppm as (CaCO <sub>3</sub> + MgCO <sub>3</sub> ) |
| Chloride                     | 40 ppm as (Cl)                                              |
| Sulfate                      | 100 ppm as (SO <sub>4</sub> )                               |

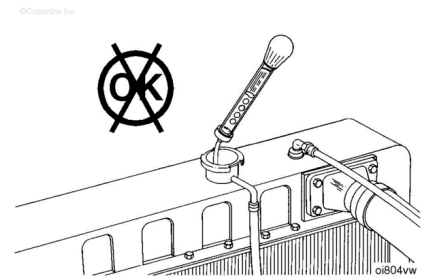
Fully formulated antifreeze **must** be mixed with quality water at a 50/50 ratio (40-percent to 60-percent working range). A 50/50 mixture of antifreeze and water gives a -36°C [-33°F] freezing point and a 110°C [230°F] boiling point, which is adequate for locations in North America. The actual lowest freezing point of ethylene glycol antifreeze is 68 percent. Using higher concentrations of antifreeze will raise the freezing point of the solution and increase the possibility of a silica gel problem.



A refractometer **must** be used to measure the freezing point of the coolant **accurately**.



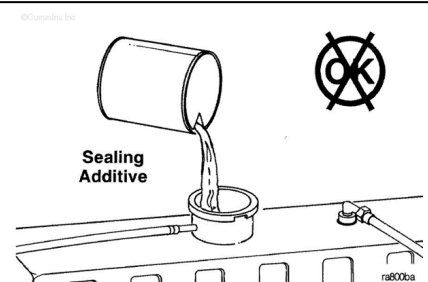
Do **not** use a floating ball hydrometer. Using a floating ball hydrometer can give an incorrect reading.



## Cooling System Sealing Additives

Do **not** use sealing additives in the cooling systems. The use of sealing additives can:

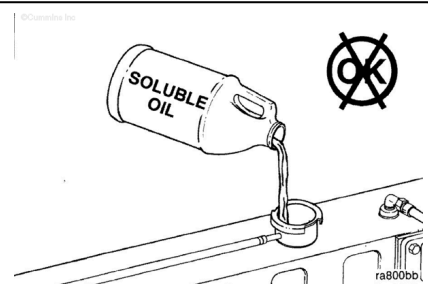
- Build up in coolant low-flow areas
- Clog coolant filters
- Plug radiator and oil cooler.



## Cooling System Soluble Oils

Do **not** use soluble oils in the cooling system. The use of soluble oils can:

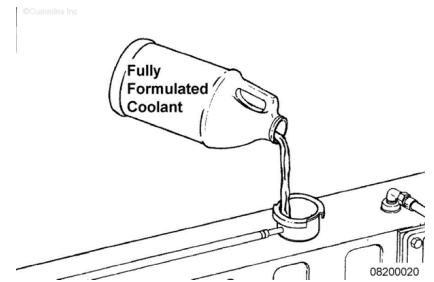
- Allow cylinder liner pitting
- Corrode brass, aluminum, and copper
- Damage heat transfer surfaces
- Damage seals and hoses.



## Supplemental Coolant Additive (SCA)

### ⚠ CAUTION ⚠

The engines use aluminum parts that are in contact with the coolant. Improper coolant, coolant filter selection, and maintenance will likely result in perforation of one of these parts. Insufficient concentration of the coolant additives will result in liner pitting and engine failure.



Fully formulated products contain SCAs and are required to protect the cooling system from scale and fouling, solder corrosion, and general corrosion. The coolant filter is required to protect the cooling system from abrasive materials, debris, and precipitated coolant additives.

## Fleetguard® DCA4 Service Filters and Liquid Precharge

Coolant Filter Options for Standard Service or Extended Service Intervals:

**WF2125** - This filter designed for use with a Fleetguard® extended service cooling system, which extends cooling system service up to 1 year, 241,402 km [150,000 miles], or 4000 hours, whichever comes first. This filter is used for cooling systems up to 76 liters [20 gal]. Refer to Service Bulletin, Coolant Requirements and Maintenance, Bulletin 3666132 (</qs3/pubsys2/xml/en/bulletin/3666132.html>). Fleetguard® ES coolant must be used for all fill and top-off, which is critical for extended cooling system maintenance intervals.

**WF2126** - This filter is designed for extended service intervals up to 80,467 km [50,000 miles] when using TMC RP329 or RP330 coolants and is used for cooling system up to 76 liters [20 gal]. Refer to Service Bulletin, Coolant Requirements and Maintenance, Bulletin 3666132 (</qs3/pubsys2/xml/en/bulletin/3666132.html>).

**WF2127** - This filter has been designed for extended service interval of 80,467 to 241,402 km [50,000 to 150,000 miles]. It has no chemical additives and can be used in the following systems:

- Cooling systems above 76 liters [20 gal] in capacity.
- See maintenance chart in this procedure.

When using WF2127 filter, the following volumes of treatment **must** be added at the designated mileages:

| Cooling System Capacity                                                                                                                                                                                                                                                             | 80,467 km [50,000 mile]<br>Service Interval with the<br>RP329/330 Coolant or<br>Treated Water (Fleetcool or<br>DCA4) | 241,402 km [150,000 mile] -<br>Service Interval with ES<br>Coolant (ES Liquid). |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 0 to 76 liters [0 to 20 gal]                                                                                                                                                                                                                                                        | 1 liter [1 qt] (10 units)                                                                                            | 1 liter [1 qt] (15 units)                                                       |
| 76 to 151 liters [20 to 40 gal]                                                                                                                                                                                                                                                     | 1.9 liter [2 qt] (20 units)                                                                                          | 1.9 liter [2 qt] qt (30 units)                                                  |
| 151 to 227 liters [40 to 60 gal]                                                                                                                                                                                                                                                    | 2.8 liters [3 qt] (30 units)                                                                                         | 2.8 liters [3 qt] (45 units)                                                    |
| 227 to 303 liters [60 to 80 gal]                                                                                                                                                                                                                                                    | 3.8 liters [4 qt] (40 units)                                                                                         | 3.8 liters [4 qt] (60 units)                                                    |
| 303 to 379 liters [80 to 100 gal]                                                                                                                                                                                                                                                   | 4.7 liters [5 qt] (50 units)                                                                                         | 4.7 liters [5 qt] (75 units)                                                    |
| <b>Note :</b> Filters <b>must</b> meet Cummins Inc. SAM 10769. Fleetguard® filters meet Cummins Inc. SAM 10769. For systems larger than 416 liters [100 gal], use 1 liter [1 qt] of SCA per 57 liters [15 gal]. Consult the vehicle manufacturer for total cooling system capacity. |                                                                                                                      |                                                                                 |

## Testing SCA Concentration Level CC-2602 Test Kit

### Precautions and Instructions for Proper Kit Use

Carefully follow the instructions to test the coolant. Take the appropriate action recommended by the kit.

- The coolant sample to be tested **must** be between 10 and 54°C [50 and 130°F]. If the sample is too cold or too hot, you will get incorrect results.
- To get the best color match results, compare test strip pads to the color chart in daylight or under cool white fluorescent lighting. If unsure about a specific color match when a test does fall between two colors on the color chart, choose the lower numbered (or lettered) block. It is safer to underestimate your results than to overestimate.
- The test strips do have a limited shelf life and are sensitive to humidity and extreme heat. Proper handling and storage is necessary to protect the life of the strips.
- Keep the cap tightly sealed on the test strip bottle **except** when removing a strip. Store away from direct sunlight and in an area where the temperature will generally stay below 32°C [90°F].
- Do **not** use the test strips after the expiration date stamped on the bottle.
- Discard the kit if the top pad on the unused strips have turned light brown.
- Use one strip at a time and take care **not** to touch any of the pads on the strip. Doing so will contaminate the pads and skew the test results.
- If the strip container is left uncapped for 24 hours, moisture in the air will render the strips useless, even though no discoloration will be evident.

- **Only** use the color chart supplied with the kit.
- Following the correct test times is very important. Use a clock or stopwatch.
- Do **not** utilize the test kit to maintain minimum SCA concentration levels (i.e., 1.5 units).
- When performing service that requires draining the cooling system, take special precautions to collect coolant in a clean non-galvanized container, seal coolant to prevent contamination, and save for reuse.

### Probablizer:

- 3318169S Plug - Installs on the engine for easy coolant sampling
- 3318168S Cap - Use with Monitor C™ bottle to sample coolant
- CC2700 Monitor C™ - Use lab analysis of coolant samples for more detailed analysis.

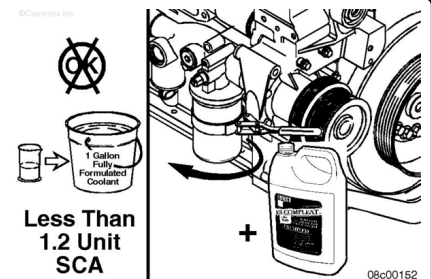
### CC2602 Coolant Test kit

- Works with any SCA formulation. Call 1-800-22FILTER (800-223-4583) if you have this test kit and the color chart does **not** show the number of units of SCA gallon of coolant.

## Test Intervals

### **⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**



Testing is recommended if the operator is **not** sure of his cooling system condition due to leaks, uncontrolled topping off of the system, or major coolant loss.

If the concentration is below 1.2 units per 3.8 liters [1 gal], replace the filter and precharge with liquid.

Check the SCA concentration level at least every 6 months, and anytime the coolant condition is unknown or corrosion is apparent within the cooling system.

Use Fleetguard® coolant test kit, CC2602, to check the concentration level. Instructions are included with the test kit.

Call the following numbers to get answers to any questions you have about cooling system maintenance:

|                       |                       |
|-----------------------|-----------------------|
| <b>Cummins Inc.</b>   | <b>Fleetguard®</b>    |
| <b>1-800-DIESELS</b>  | <b>1-800-22FILTER</b> |
| <b>1-800-343-7357</b> | <b>1-800-223-4583</b> |

## Coolant Replacement Requirements

### **⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**

### **⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

**Note :** Dispose of used coolant/antifreeze in accordance with federal, state, and local laws and regulations.

Drain and flush the cooling system after 6000 hours or 3 years of service. However, if Fleetguard® ES coolant and ES filters are used, check chloride, sulfate, and pH levels according to Service Bulletin, Coolant Requirements and Maintenance, Bulletin 3666132

(</qs3/pubsys2/xml/en/bulletin/3666132.html>), to determine whether the coolant **must** be replaced. Refill with either new fully formulated coolant or ES coolant.

